

Amanda Bevan Zientek, PhD

Director, Biodiversity Resources Center, Hudsonia

Professional Summary: My professional career has focused on: 1) improving educational experiences in higher education, 2) mentoring undergraduate students, and 3) engaging with my community in the fields of ecology, natural resource management, and conservation. I am passionate about improving access to transformative experiences in education for learners of all ages, and I work to integrate inclusive practices that welcome students to experience science and increase community involvement. My goal is to promote the building of life-long scientific inquiry skills and demonstrate the joy of learning.

Education

Ph.D. in Biology (2023)

University of Northern Colorado

School of Biological Sciences

Advisor: Dr. Rick Adams

Dissertation: Response of bats and nocturnal food webs to mountain pine beetle (*Dendroctonus ponderosae*) outbreaks.

B.S. in Ecology, Evolution, and Natural Resources (2016; *cum laude*)

Rutgers University

School of Biological and Environmental Sciences

Advisor: Dr. Brooke Maslo

Honors thesis: Survival analysis of Indiana bats (*Myotis sodalis*) with white-nose syndrome

Teaching Experience

Visiting Assistant Professor

Fall 2024

Department of Biology, Bard College

- Developed and taught the 100-level mammalogy lecture and lab course *Exploring Mammalogy: An Introduction to Scientific Inquiry* (BIO160) centered on building scientific inquiry skills
- Used open-access resources following the [C.R.E.A.T.E.S method](#) to improve scientific literacy through the critical, scaffolded review of peer-reviewed literature
- Integrated a Course-based Undergraduate Research Experience (CURE) as a core component of the course which included the full cycle of student-developed, independent research of mammals on Bard's campus (proposal development to scientific poster presentation)

Instructor of Record

Fall 2022

School of Biological Sciences, University of Northern Colorado

- Developed and taught the 300-level biology lecture and lab course *Ecology* (BIO360) employing active-learning techniques including learner-centered group activities analyzing publicly available datasets and primary literature

- Collaborated with City of Greeley's Environmental Planner to develop a Course-based Undergraduate Research Experience (CURE) for post-restoration river monitoring as a core component of the lab
- Coordinated scientific poster session for undergraduate students in BIO360, enabling students to present research results to city officials and peers

Human Anatomy Lab (BIO245) Coordinator

Fall 2019 – Spring 2023

School of Biological Sciences, University of Northern Colorado

- Created introductory videos, lab activities, quizzes, and practicals for 14 Human Anatomy Lab sections for 8 semesters following a flipped classroom model (including virtual assignments throughout the Covid-19 pandemic)
- Led weekly meetings to train graduate and undergraduate teaching assistants on course material
- Prepared and maintained course platforms (Canvas) for all lab sections

Ecology (BIO360) Graduate Teaching Assistant

Fall 2021 – Spring 2022

School of Biological Sciences, University of Northern Colorado

- Assisted with lecture activities and created learner-centered materials for lecture including virtual assignments

Human Anatomy (BIO245) Graduate Teaching Assistant

Fall 2019 – Spring 2023

School of Biological Sciences, University of Northern Colorado

- Taught one to three sections of Human Anatomy Laboratory (BIO245) for 18 to 24 students
- Assisted with case study activities as a lecture assistant and graded assignments

Biology: Atoms to Cells (BIO110) Graduate Teaching Assistant

Fall 2018 – Spring 2019

School of Biological Sciences, University of Northern Colorado

- Taught two lab sections for an introductory biology course for biology majors which used a semester-long Course-based Undergraduate Research Experience (CURE)
- Assisted with group-based activities as a lecture assistant and graded assignments

Exploring Biology Lab (BIO105) Graduate Teaching Assistant

Fall 2018 – Spring 2019

- Taught a lab section for an introductory biology course for non-majors
- Planned lessons and assessments and graded quizzes

Education Specialist

September 2016 – February 2018

Organization for Bat Conservation

- Designed and delivered school-based (K-12) and public outreach programs on ecological topics, engaging groups of 20-400 people of all ages

Pedagogical Training & Coursework**University of Northern Colorado**

- Foundations in Equity and Inclusion (2021)
- Equity Minded Syllabus (2021)
- Teaching College Biology (BIO631) (Spring 2019)
- College Biology Course Development (BIO632) (Spring 2020)

Rutgers University

- Project Learning Tree (2015)
- Project W.E.T (Water Education for Teachers) (2015)

Research & Professional Experience

Biodiversity Resources Center, Hudsonia Ltd., NY

October 2023 – present

Director

- Lead Hudsonia's Biodiversity Resources Center, overseeing outreach and continuing education workshops focusing on ecology and conservation of natural resources for land-use decision-makers in partnership with Bard College and Cornell University
- Study impacts of land management and farming practices on tri-colored bat (*Perimyotis subflavus*) space use and activity patterns
- Oversee staff and undergraduate student interns conducting field surveys and manage datasets and grant budgets
- Partner with Environmental Justice Fellows, a youth-led initiative, to identify important greenspaces in Newburgh, NY
- Prepare proposals for competitive grant funding from state and non-governmental agencies to conduct ecological monitoring surveys of bats, birds, herpetofauna, and rare plants, and prepared Natural Resource Inventories for municipal agencies
- Design and conduct field surveys to identify high-priority habitats to inform ecological restoration, management, and monitoring programs
- Develop wildlife and habitat management plans in collaboration with non-governmental organizations and landowners
- Investigate the impact of hemlock woolly adelgid (*Adelges tsugae*) on forest composition and structure using a long-term ecological dataset and multivariate analyses
- Conduct geospatial analyses in ArcGIS Pro and create maps for Natural Resource Inventories and biodiversity surveys for technical and non-technical audiences

Adams Bat Lab, University of Northern Colorado, CO

August 2018 – May 2023

Independent & Collaborative Researcher

- Quantified bat assemblage structure in bark beetle impacted forests in the Northern Front Range of Colorado using acoustic surveys
- Conducted multivariate analyses to describe impacts of disturbance on the forest community and identify food web variables (insect prey availability and habitat structure) associated with bat activity patterns
- Wrote conservation plan for tri-colored bats (*Perimyotis subflavus*) for Boulder County Parks and Open Space (BCPO)
- Quantified large mammal use of trails and drinking holes in arid environments using camera traps
- Processed and analyzed acoustic data to model tri-colored bat responses to novel habitat features using nonparametric multiplicative regression models
- Conducted mist net surveys to quantify seasonal bat activity patterns in BCPO parks and to monitor changes in population structure from year-to-year

Holt Lab, University of Northern Colorado, CO

January 2019 – May 2023

Collaborative Researcher

- Created and evaluated a classroom intervention to broaden undergraduate perceptions of colonialism, sexism, and racism in the field of ecology
- Conducted analyses using machine learning (e.g. conditional random forest models) to determine best predictors of perceived barriers to biology doctoral student success from survey data
- Identified and adapted instruments from peer-reviewed literature and dissertations for questionnaire survey to quantify perceived barriers for biology doctoral students

Rocky Mountain Amphibian Project, Natural Diversity Database, WY May 2018 – October 2018*Project Manager*

- Recruited, trained, and supervised community participants on amphibian survey methods and project goals
- Radio tracked boreal toads and characterized diurnal activity patterns and habitat features using vegetation surveys to determine why this species has low fungal loads of chytrid fungus (*Batrachochytrium dendrobatidis*) compared to other amphibian species in the region
- Managed project budget and oversaw equipment inventory
- Coordinated partnerships with Wyoming Game and Fish Department and U.S. Forest Service

Organization for Bat Conservation, Pontiac, MI

September 2016 – February 2018

Urban Bat Project Lead and Education Specialist

- Collaborated with partners in ten cities across U.S. to build, install, and distribute bat houses, plant “bat-friendly” gardens, and organize volunteer-led bat walks
- Trained partners and community members to use acoustic monitoring equipment
- Organized and led educational bat walks in urban environments
- Developed quantitative and qualitative surveys to assess how participatory science projects may impact participant attitudes towards conservation

Wyoming Natural Diversity Database, WY

Summer 2016

Bat Field Technician

- Conducted mist net surveys and acoustic monitoring surveys
- Radio tracked bats to roosts in steep, rocky terrain
- Performed vegetation surveys and collected habitat data to characterize roost site selection

Youth Advisory Board, Rutgers University, NJ

January 2014 – May 2016

Central Region Ambassador

- Facilitated discussion groups of young people ages 12-23 to share experiences with foster care and juvenile justice system
- Worked with youth and community partners to investigate/promote solutions to improve state care

New Jersey Non-Game Endangered Species Program, NJ

February 2015 – August 2015

Research Assistant

- Conducted mist net surveys and radio-tracked northern long-eared bats to roost sites
- Surveyed bats in bat houses in suburban areas as part of long-term monitoring effort

Research Mentorship Experience

Summary: Below are students that I have mentored and the titles of their research projects. This process included advising students on topic development, securing funding through written proposals, preparing for field work or interviews, summarizing and presenting final results in a scientific poster at conferences, and preparing and submitting manuscripts to peer-reviewed journals*

*Denotes manuscript in progress in coordination with an undergraduate student; see manuscript section for more details

Bard College

Jenna Holub

July 2025 – present

- Municipal officials' awareness and understanding of threats to forests and available tools for forest conservation in the Hudson Valley

Bates College

*Yoko Franklin**

June 2024 – September 2024

- Activity of eastern red bats in riparian habitats is greater near non-conventional, organic farms than in nature preserves

University of Northern Colorado

*Ashley Trump**

Fall 2022 – Summer 2023

- Estimating primary productivity following severe bark beetle outbreaks using dendrochronology

*Alexandria Colpitts**

Summer 2021

- Describing tri-colored bat (*Perimyotis subflavus*) roosting ecology in novel habitats

Lucas Allen

Summer 2021

- Examining the relationship between bat body size condition and ectoparasite load

Tara Hobbs

Summer 2020

- Relating bat ecomorphology and drinking activity at high elevation ponds

Donovan Barret

Summer 2020

- Quantifying the impacts of bark beetle kill on pond water quality

Jenny Shokar

Summer 2019

- Analyzing acoustic traits of western long-eared bat (*Myotis evotis*) in beetle-killed forests

Morgan Marzulla

Summer 2019

- Quantifying changes in moisture content and litter layer following bark beetle kill

Colorado State University

*Charli Geer**

Summer 2022 – Fall 2022

- Identifying nocturnal forest insects in beetle-killed forests

Peer Reviewed Publications*

*Indicates undergraduate student that I mentored directly

Sanchez K., **Bevan Zientek A.**, Holt E. 2025. Social justice awareness in the undergraduate ecology classroom. *Ecosphere* 16(3): e70151.

Bevan Zientek A., Colpitts A.*, Adams, R.A. 2024. Forest characteristics predict tricolored bat activity within novel Colorado habitats. *Journal of Mammalogy* 105(5): 1057-1066.

Sanchez K., **Bevan A.**, Vita A., Royse E., Januszkiewicz E., Holt E. 2023. Grit matters: Passion and perseverance predict doctoral student perception of barriers. *Journal of Biological Education* 58(5): 1262-1285.

Maslo B., Stringham O., **Bevan A.**, Brumbaugh A., Sanders C., Hall M., Fefferman N. 2017. High annual survival in infected wildlife populations may veil a persistent extinction risk from disease. *Ecosphere* 8(12): e02001.

Maslo B., Valentin R., Leu K., Kerwin K., Hamilton G.C., **Bevan A.**, Fefferman N.H., Fonseca D.M. 2017. *Chiro-surveillance*: the use of native bats to detect invasive agricultural pests. *PLOS ONE* 12(3): e0173321.

Manuscripts In Progress*

*Indicates undergraduate student that I mentored directly

Adams, R. A., **Bevan Zientek, A.** New Horizons: Tricolored bats (*Perimyotis subflavus*) show increasing numbers at two wildlife areas in Colorado. (*Submitted – Western North American Naturalist*).

Bevan Zientek, A., Franklin, Y*. Activity of eastern red bats in riparian habitats is greater near non-conventional, organic farms than in nature preserves (*Submitted – Journal of North American Bat Research*).

Benedict L., **Bevan Zientek A.** Trill rate and frequency are optimized for long distance transmission for long songs (*In prep*).

Bevan Zientek, A., Trump*, A., Geer*, C., Franklin, S. Insect abundance and diversity increase with changes in understory plant community structure following bark beetle outbreaks (*In prep*).

Selected Technical Reports

Bevan Zientek, A., and Leonardi, A. 2025. Monitoring Plan for Biodiversity Conservation at Sky High Farm. Prepared for Sky High Farm.

Bevan Zientek, A. 2025. Preliminary Survey of Tri-colored Bat Use of Farm Hub Riparian Habitats. Prepared for Farm Hub.

Bevan Zientek, A. 2024. Biodiversity Assessment of the Quassaick Creek Greenway. Prepared for Scenic Hudson.

Bevan Zientek, A., Graham, C., Stevens, G., Vispo, C., Knab-Vispo, C., Bradford, K., Tesauero, J., Schmidt, B., Federman, L., and White, E. 2024. Biological and Water Quality Surveys at the Jesse and Gayle Bontecou Wildlife Conservation Preserve. Prepared for Dutchess Land Conservancy.

Adams, R. A., and **Bevan, A.** 2018. Continued Survey of Bats in Boulder County. Prepared for Boulder County Parks and Open Space.

Bevan, A. and Adams, R.A. 2019. Boulder County Conservation Management Plan for the Tri-colored Bat (*Perimyotis subflavus*). Prepared for Boulder County Parks and Open Space.

Invited & Conference Presentations*

*Received award for presentation. See Awards & Honors section

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- Bevan Zientek, A.** 2024. Bioacoustics and Bats: Unraveling the Ecology of Elusive and Imperiled Organisms (**Talk**). Bard Biology Program Seminar Series, Annandale, NY.
- Bevan, A., Adams, R. A.** 2022. *Response of bats to severe bark beetle outbreaks and ensuing successional changes* (**Talk**). Annual meeting of the Ecological Society of America & Canadian Society for Ecology and Evolution joint meeting, Montreal, Quebec, Canada.
- Bevan, A., Sanchez, K., Holt, E.** 2022. *Class intervention to address representation and social justice issues in ecology* (**Poster Presentation**). Ecological Society of America & Canadian Society for Ecology and Evolution joint meeting, Montreal, Quebec, Canada.
- Bevan, A.*, Adams, R. A.** 2022. *Response of bats to severe bark beetle outbreaks and ensuing successional changes* (**Talk**). North American Symposium for Bat Researchers & the International Bat Researchers Conference joint meeting, Austin, TX.
- Bevan, A.*, Adams, R. A.** 2022. *Response of bats to severe bark beetle outbreaks and ensuing successional changes* (**Talk**). Front Range Student Ecology Symposium, Colorado State University, Fort Collins, CO.
- Bevan, A.** 2021. *The response of bats and associated nocturnal food webs to bark beetle kill* (**Talk**). Soup and Science, Natural Resources and Ecology Lab, Colorado State University, Fort Collins, CO.
- Bevan, A., R. A. Adams.** 2019. *Response of Bats and Nocturnal Food Webs to Mountain Pine Beetle Outbreaks* (**Poster Presentation**). North American Symposium for Bat Research. Kalamazoo, MI.
- Bevan, A, G. Auterie.** 2017. *Does the framework of citizen science projects affect participants' attitudes towards conservation?* (**Poster Presentation**). North American Symposium for Bat Research., Knoxville, TN.
- Kerwin, K., L. Tsen, M. Arias, **A. Bevan**, B. Schum, B. Maslo. 2016. *A systematic review of roost-site characteristics of forest-dwelling bats affected by white-nose syndrome* (**Poster Presentation**). Annual meeting of the Northeast Fish & Wildlife Association. Annapolis, MD.
- Maslo, B., R. Valentin, K. Leu, K. Kerwin, **A. Bevan**, G.C. Hamilton, N. H. Fefferman, D. M. Fonseca. 2015. *Chiro-surveillance: the use of native bats to detect invasive agricultural pests* (**Poster Presentation**). Annual meeting of the Ecological Society of America. Baltimore, MD.

Community Outreach Presentations

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- Bats - NY's Most Threatened Mammal, Morton Memorial Library, Rhinecliff, NY (2024)
 - Bat Walk Community Program. Bontecou Preserve, Dutchess Land Conservancy, NY (2024)
 - Bats of Larimer County. City of Fort Collins Natural Areas Department, CO (2022)
 - Bat Walk Community Program. City of Fort Collins Natural Areas Department, CO (2022)
 - Bat Chat: Factors influencing bat activity in Red Feather Lakes Wildlife Committee for Crystal Lakes Road and Recreation Association, CO (2022)
 - Bat Walk Community Program. City of Fort Collins Natural Areas Department, CO (2021)
 - Bat Conservation Proposal for the West Greeley Conservation District (2020)
 - Professional Development in Ecology. Department of Ecology, Evolution, and Natural Resources course, Rutgers University (2020)

- Bats in Roosevelt National Forest Near You! Crystal Lakes Recreation Department, Monthly Wildlife Committee meeting, CO (2019)
- Educating, Inspiring, and Activating Urban Communities to Save Bats. Annual meeting of the North American Society for Bat Research, Teacher Workshop (2017)
- Bat Appreciation Day- Live Chat with Amanda Bevan. Explore.org (2017)
- Studying Bats at Devils Tower National Monument. Devils Tower National Monument (2016)
- Backyard Wildlife Management. Master Gardener Association of Passaic County, NJ (2016)

Awards & Honors

- Dean's Citation Award for Dissertation Excellence, University of Northern Colorado (2023)
 - Only 2 awarded each year
- Natural and Health Sciences Academic Scholar Award, University of Northern Colorado (2023)
- Wildlife Acoustics Science Award, International Bat Researchers Conference and North American Society for Bat Researchers (2022)
 - Award for oral presentation at joint conference with only 10 awarded each year
- Front Range Student Ecology Symposium, Outstanding Presentation (2022)
- National Science Foundation Graduate Research Fellowship Program, *Honorable Mention* (2020)
- Dr. Applegate Award for Outstanding Student in Wildlife Conservation, Rutgers University (2016)

Grants, Fellowships, & Scholarships

- Hudson River Estuary Program, Local Stewardship Planning Grants – \$127,000 (2024 - 2025)
- Animal Behavior Society Graduate Student Travel Award – \$700 (2023)
- Natural and Health Science Student Research Fund, University of Northern Colorado – \$500 (2022)
- Graduate Student Association Conference Grant, University of Northern Colorado – \$380 (2022)
- Graduate Student Association Research Grant, University of Northern Colorado – \$1,300 (2021)
- Denver Audubon Society's Lois Weber Fund Grant – \$1,637 (2021)
- Natural and Health Science Student Research Fund, University of Northern Colorado – \$500 (2021)
- Dr. Winchester Graduate Fellowship in Biology, University of Northern Colorado – \$762 (2021)
- Dr. Winchester Graduate Fellowship in Biology, University of Northern Colorado – \$784 (2020)
- Graduate Student Association Research Grant, University of Northern Colorado – \$2,055 (2020)
- Graduate Student Association Research Grant, University of Northern Colorado – \$1,163 (2020)
- Graduate Student Association Research Grant, University of Northern Colorado – \$580 (2019)
- Wildlife Acoustics Research Equipment Grant – \$1,200 value (2017)
- Aresty Conference Funding Grant, Rutgers University – \$500 (2016)
- Center for Global Education Scholarship, Rutgers University – \$1,500 (2016)
- Cook Community Alumni Association Scholarship, Rutgers University – \$1,000 (2016)
- Department of Ecology, Evolution, and Natural Resources Grant, Rutgers University – \$300 (2016)
- School of Environmental and Biological Sciences Grant, Rutgers University – \$300 (2016)
- Center for Global Education Scholarship, Rutgers University – \$2000 (2014)

- Price Family Fellowship – \$5500 (2014 – 2016)
- Rutgers National Scholarship, Rutgers University – \$32,000 (2011 – 2015)

Selected Popular Press Interviews

Radio Kingston, Nature Nuggets. Episode 8 ~ Duane Martinez with Amanda Bevan Zientek. Radio interview (June 08, 2024)

BeProvided Conservation Radio. Amanda Bevan: Educator for Organization for Bat Conservation. Podcast interview (October 31, 2017)

Smithsonian Magazine. How a Deadly Flesh-Eating Fungus Helped Make Bats Cute Again. Article interview (October 26, 2017)

The Mid-Michigan Pet Expert Talk Show. Bats and Black Cats. Radio interview (October 16, 2017)

Stateside-Michigan Radio, National Public Radio. As Bat Populations Decline, Conservationists Urge Urbanites to Welcome New Nocturnal Neighbors. Radio interview (September 2017)

The DIRT-American Society of Landscape Architects. Are Urban Bats the Future? Article interview (June 30, 2017)

Service

- Reviewer, *Acta Chiropterologica* (May 2023 – October 2023)
- Board Member, North American Bat Researchers Symposium (August 2022 – October 2024)
- Teacher Appreciation Resource Reception, West Greeley Conservation District (June 2021)
- Board member, Colorado Bat Society (Fall 2018 – present)
- Created outreach materials on bats for West Greeley Conservation District (Spring 2019)
- Panel member, McNair Scholar Program (Spring 2019)
- Endangered Wyoming Toad Release, WY Game and Fish Department (June 2018)
- English Instructor, Liceu Schools, Brazil (February 2018 – May 2018)
- Research Assistant, African Bat Conservation, Malawi (January 2016)

Skills

Programming	R, Python
Statistics	Multivariate analyses (PCA/NMS, structural equation modeling), data wrangling, data visualization, survey design and analysis (including coding)
Software	GIS (ArcGIS, QGIS), R, Git, Google suite, Microsoft suite
Field Techniques	Acoustic analysis, mist netting, small mammal trapping, radio telemetry, bat identification, vegetation surveys, habitat mapping, insect sampling

Certifications & Other Training

- Introduction to Bayesian Modeling, Ecological Society of America (2024)
- Winter Tree & Shrub Identification, Eagle Hill Institute (2023)
- Spatial Ecology: GIS Analysis in R, Smithsonian-Mason School of Conservation (2021)

- Multivariate Analysis of Community Data in PC-ORD, University of Northern Colorado (Spring 2019)
- CPR, American Red Cross (October 2017)
- Environmental Geomatics (GIS), Rutgers University (2016)